

Dynamic Pricing and Demand Forecasting

A Machine Learning Approach to Pricing Strategy Optimization

Muhammad Fiaz

Domain: Pricing Strategies | Marketing Research

Dataset: Store Item Demand Forecasting Challenge — Kaggle
2026

1. Project Overview

This project applies machine learning techniques to retail demand forecasting and dynamic pricing strategy evaluation. Using five years of daily sales data from the Kaggle Store Item Demand Forecasting Challenge, the project builds and compares multiple forecasting models, simulates three pricing strategies, and derives optimal pricing and budget allocation recommendations.

The central research question is: Can machine learning models accurately forecast item-level demand well enough to support real-time dynamic pricing decisions, and what are the implications of myopic versus long-term pricing strategies on revenue?

1.1 Dataset Description

Column	Type	Description
date	datetime64	Date of sale from 2013-01-01 to 2017-12-31
store	int64	Store identifier (1 to 10)
item	int64	Item identifier (1 to 50)
sales	int64	Units sold on that date

- Total Records: 913,000
- Stores: 10 | Items: 50
- Time Period: 5 years with daily granularity (2013 to 2017)
- No missing values, no holiday effects, no store closures

1.2 Tools and Technologies

Category	Tools
Language	Python 3.12
Environment	Google Colab
Data Manipulation	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn (Linear Regression, Random Forest, Gradient Boosting)
Time Series	Prophet (Meta/Facebook)
Optimization	SciPy

2. Exploratory Data Analysis

Before building models, the data was explored across four dimensions: overall trend, annual seasonality, weekly seasonality, and store and item level performance.

2.1 Overall Sales Trend

- Total daily sales show a clear upward trend from 2013 to 2017.
- Sales grew from around 13,000 units per day in early 2013 to peaks of 45,000 units per day by 2017.
- Strong weekly oscillations are visible throughout indicating consistent weekly demand cycles.

2.2 Monthly and Annual Seasonality

- All five years follow the same seasonal shape confirming a strong and predictable annual pattern.
- July is consistently the peak month across all years with average sales reaching up to 75 units per day.
- December is consistently the lowest month with sales as low as 33 units per day in 2013.
- Year-over-year growth is visible with 2017 being the highest performing year overall.

2.3 Weekly Seasonality

- Sunday is the highest sales day with an average of 62 units.
- Monday is the lowest at 41 units — roughly 33% lower than Sunday.
- Sales steadily increase from Monday through to Sunday every week.

2.4 Store and Item Performance

Finding	Detail
Top Store	Store 2 averaging 67 units per day
Weakest Store	Store 7 averaging 37 units per day
Top Items	Item 15 and Item 28 averaging 88 units per day
Weakest Item	Item 5 averaging 18 units per day
Sales Distribution	Right-skewed with skewness of 0.87, mean 52.25, median 47.00

3. Feature Engineering

Seventeen features were engineered from the original four columns to give machine learning models the temporal context needed to learn demand patterns.

Feature Type	Features Created	Purpose
Time Features	year, month, day, day_of_week, week, quarter, is_weekend	Capture calendar-based seasonality
Lag Features	lag_7, lag_14, lag_28, lag_365	Give the model memory of past sales
Rolling Features	rolling_mean_7, rolling_mean_30, rolling_std_7, rolling_std_30	Capture recent demand trends and volatility

- Data was sorted by store, item, and date before feature creation.
- 182,500 rows with NaN values were dropped due to lag and rolling window requirements.
- Final clean dataset: 730,500 rows with 19 columns and zero missing values.

4. Dynamic Pricing Simulation

Since the dataset does not contain price data, a price elasticity of demand framework was simulated. This is standard practice in marketing research when transaction-level price data is unavailable. The simulation was applied to Store 2, Item 15 across the full five-year period.

4.1 Assumptions

Parameter	Value
Base Price	10.00 USD
Price Elasticity of Demand	-1.5
High Demand Price Adjustment	+20% on days above average demand
Low Demand Price Adjustment	-10% on days below average demand

4.2 Results

Strategy	Total Revenue	vs Static
Static Pricing	1,712,170 USD	Baseline
Myopic Pricing	1,663,544 USD	-2.84%
Dynamic Pricing	1,652,951 USD	-3.46%

Both variable pricing strategies underperformed static pricing. With an elasticity of -1.5, raising prices on high demand days reduces quantity sold enough to cancel out the revenue gain from the higher price. This confirms that dynamic pricing only outperforms static pricing when demand is inelastic.

5. Demand Forecasting Models

5.1 Train/Test Split

Split	Period	Records
Training	2014 to 2016	548,000
Test	2017 (held out)	182,500

A time-based split was used to simulate a real forecasting scenario and prevent data leakage.

5.2 Models

- Linear Regression: baseline model to set a performance floor.
- Random Forest: 50 estimators, max depth 10, trained on all 500 store-item combinations.
- Gradient Boosting: 100 estimators, learning rate 0.1, max depth 5.
- Prophet: trained on Store 2 Item 15 only with multiplicative yearly and weekly seasonality.

6. Model Evaluation

Model	MAE	RMSE	R2 Score
Linear Regression	6.8779	8.9676	0.9192
Random Forest	6.4860	8.4750	0.9279
Gradient Boosting	6.2046	8.0602	0.9347
Prophet	10.5694	13.1083	0.8102

- Gradient Boosting is the best model with R2 of 0.9347 on unseen 2017 data.
- All ML models significantly outperform Prophet by leveraging 17 engineered features.
- Prophet achieved R2 of 0.81 with zero feature engineering making it practical for quick forecasts.

6.1 Top Feature Importances

Feature	Importance Score	Interpretation
rolling_mean_7	0.4886	7-day rolling average of past sales
lag_7	0.3376	Sales exactly 7 days ago
rolling_mean_30	0.0629	30-day rolling average

lag_14	0.0444	Sales exactly 14 days ago
day_of_week	0.0261	Day of the week (weekly seasonality)

rolling_mean_7 and lag_7 together account for 82% of predictive power confirming that recent sales history is the strongest driver of future demand.

7. Budget and Price Optimization

7.1 Optimal Price

Using SciPy `minimize_scalar` with a price range of 8.00 to 15.00 USD and elasticity of -1.5, the optimal daily price was computed for Store 2 Item 15 based on Gradient Boosting forecasts.

Metric	Value
Average Optimal Price	8.00 USD
Total Optimal Revenue (2017)	515,950 USD
Total Static Revenue (2017)	461,480 USD
Revenue Gain	+54,470 USD (+11.8%)

- The optimizer recommends 8.00 USD consistently because under elastic demand lowering price always increases total revenue.
- Optimal pricing generated 11.8% more revenue than static pricing in 2017.

7.2 Marketing Budget Allocation

A diminishing returns model was used to allocate a 10,000 USD budget across 10 stores to maximize total predicted sales lift.

Store	Allocated Budget (USD)	Expected Sales Lift (Units)
Store 2	1,277	129.3
Store 8	1,225	123.7
Store 3	1,140	114.6
Store 7 (lowest)	699	68.2
Total (all stores)	10,000	998.85

- Budget was proportionally allocated based on base demand — stronger stores receive more investment.
- Total expected sales lift from 10,000 USD budget is 998.85 units across all 10 stores.

8. Project Summary

Step	Description	Key Output
1. Data Loading	Kaggle API download	913,000 clean records
2. EDA	Trend and seasonality analysis	July peak, Sunday peak, Store 2 leads
3. Feature Engineering	17 features from 4 columns	730,500 rows for modeling
4. Pricing Simulation	Three pricing strategies	Static pricing wins under elastic demand
5. Forecasting Models	4 models trained	Gradient Boosting R2 = 0.9347
6. Model Evaluation	MAE, RMSE, R2 comparison	Gradient Boosting is best model
7. Optimization	Price and budget optimization	8 USD optimal, 998 units lift

Key Findings

- Gradient Boosting is the best demand forecasting model with an R2 of 0.9347 on unseen 2017 data.
- rolling_mean_7 and lag_7 account for 82% of predictive power — recent history is the strongest demand signal.
- Sunday is the peak sales day and July is the peak month consistently across all five years.
- Under elastic demand (PED = -1.5), variable pricing underperforms static pricing by up to 3.46%.
- The optimal price for Store 2 Item 15 is 8.00 USD generating 11.8% more revenue than static pricing.
- Marketing budget allocation using diminishing returns produces 998.85 units of total sales lift from 10,000 USD.